

AAYUSH KUMAR

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Education

Vellore Institute of Technology

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Vellore, Tamil Nadu

Jun 2024 – May 2028 (Expected)

Technical Skills

Languages: Python, Go, JavaScript, SQL, C++

Frameworks & Libraries: FastAPI, Django, Flask, React.js, Next.js, Express.js, Node.js, LangGraph

Tools & Technologies: Docker, Kubernetes, Linux, PostgreSQL, MongoDB, Redis, RabbitMQ, S3, OCI, ffmpeg, Nginx, Git

Experience

Loopedin

Dec 2025 – Present

Software Engineer Intern

Remote

- Built core systems for **IdeaWave**, an AI-native collaborative canvas for planning, diagramming, code snippets, comments, files, and agent-driven artifact generation.
- Implemented a typed board-state architecture for **7+ reusable primitives**, enabling consistent selection, mutation, linking, persistence, and realtime collaboration across notes, nodes, edges, code cards, context cards, comments, and agent cards.
- Helped Develop MCP/ACP-based agent workflows and S3-backed context cards, allowing CLI-connected agents to consume selected board context, spawn primitives, generate flowcharts, write code cards, and update editable board artifacts.

Yuusoft

Jul 2025 – Aug 2025

Software Engineer Intern

Remote, Singapore

- Engineered an AI-assisted automation tool using Discord + Claude to modify code, generate patches, and push Git commits via natural language, significantly improving developer workflow efficiency.
- Built core backend and platform features for Yuusoft's visual novel creator tool, including scene-management logic, input-processing pipelines, and structured data flows.
- Improved platform performance and responsiveness by optimizing state management and render paths, reducing UI latency and unnecessary re-renders by 30% and improving overall UX reliability.

Projects

Rendition - Distributed Video Transcoding System | Python, FastAPI, RabbitMQ, S3, ffmpeg, Next.js | 🐙 [GitHub](#)

- Built a distributed video transcoding platform that streams large browser uploads directly to S3-compatible storage via presigned multipart uploads and generates adaptive HLS renditions using horizontally scalable **ffmpeg** workers.
- Engineered a fault-tolerant job orchestration system with PostgreSQL, RabbitMQ, and the Outbox pattern, implementing worker leasing, heartbeats, stale-job recovery, retry backoff, and dead-letter handling to prevent lost or duplicate work under failures.
- Implemented source-aware encoding, master-playlist generation, and API-mediated playback delivery, alongside a Next.js dashboard providing upload progress tracking, job control, and HLS video playback.

Crate - Daemonless Container Runtime | Go, Linux Namespaces, OCI Images | 🐙 [GitHub](#)

- Built a daemonless container runtime in Go supporting both rootless and rootful containers, with runtime state persisted directly on disk instead of a long-lived background daemon.
- Implemented container isolation using Linux namespaces (PID, UTS, Mount, User, Network), **pivot_root**-based filesystem setup, rootless networking via **pasta**, and PTY-backed interactive process execution.
- Developed OCI/Docker image support including registry pulls, manifest resolution, layer extraction, local blob/config storage, detached container lifecycle management (**start**, **stop**, **logs**, **ps**), and reference-based blob pruning.

Kargo - Open-Source PaaS on Kubernetes | Express.js, Next.js, MongoDB, Docker, Kubernetes | 🐙 [GitHub](#)

- Developed core backend services for an open-source PaaS platform that automatically deploys user-provided Docker images onto Kubernetes, building secure APIs for authentication, deployment orchestration, and lifecycle management.
- Engineered automatic Kubernetes manifest generation (Deployments, Services, Ingress, volumes, secrets), eliminating manual cluster interaction and cutting deployment setup time by 60%.
- Improved deployment stability, scalability, and observability by adding container-status monitoring, error-resilient rollout logic, and health-check mechanisms to ensure a reliable and production-grade application.

Achievements & Extracurricular

- Google Summer of Code 2025 @AboutCode:** Contributed to the open-source ecosystem by building a high-performance file explorer for [ScanCode.io](#), improving navigation across large license, copyright, and vulnerability scan results.
- Patent Published - Encrypted Traffic Intrusion Detection (2026):** Co-invented a multi-modal AI cybersecurity system for detecting threats in encrypted network traffic using metadata analysis and anomaly detection models.
- Volunteer - Gravitas Website Team:** Helped build the backend for VIT's annual technical fest platform, serving **65k+ unique users** and handling **1M+ page visits**.
- Winner - Cicada 3301 (CTF Competition):** Secured **1st place** in a university-wide cybersecurity challenge.
- Third Place - Coderush 2.0 (Competitive Coding):** Achieved **3rd place** in a university-wide coding competition.